MIL-S-19500/171B(NAVY)
20 November 1968
SUPERSEDING
MIL-S-19500/171A(NAVY)
13 October 1961

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, GERMANIUM, TYPE 2N1853

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a germanium, PNP switching transistor and is in accordance with MIL-S-19500, except as otherwise specified herein.

1.2 Physical dimensions. See figure 1 (TO-5).

1.3 Maximum ratings.

I_C	P_T 1/	P_{EB} 2/	T_{stg}	V_{CB}	V_{EB}	V_{CE} (R_B=1KΩ)
mAdc	mW	mW	°C	Vdc	Vdc	Vdc
-100	150	5	-55 to +85	-18	-4	-17

1/ Derate linearly 25 mW/° C for T_A > 30° C.
2/ Derate linearly 5/9 mW/° C for T_A > 25° C.
This rating refers to operation of the emitter base diode in the breakdown region.

1.4 Primary electrical characteristics.

Parameter	I_{CBO}	h_{FE}	V_{CE} (sat)
Conditions	V_{CB} = -18 Vdc T_A = +60° C	V_{CE} = -1 Vdc I_B = -0.20 mAdc	I_C = -6 mAdc I_B = -0.2 mAdc
Units	μAdc	---	Vdc
Minimum	---	30	---
Maximum	-35	400	-0.20

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY
MIL-S-19500 - Semiconductor Devices, General Specification for

STANDARDS

MILITARY
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts
MIL-STD-750 - Test Methods for Semiconductor Devices

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. Transistor shall be of the design, construction, and physical dimensions shown on figure 1.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I and II.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

(a) Country of origin
(b) Manufacturer's designating symbol

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I and II.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of groups A and B inspections.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

TABLE I. Group A inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			10				
Visual and mechanical inspection	2071			---	---	---	---
Subgroup 2			5				
Collector to base cutoff current	3036	T_A = +60° C Bias cond. D V_{CB} = -18 Vdc		I_{CBO}	---	-35	µAdc

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TABLE I. Group A inspection (Continued)

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 2 (Cont.)							
Collector to emitter voltage (saturated)	3071	I_C = -6.0 mAdc I_B = -0.20 mAdc		$V_{CE}(sat)$	---	-0.2	Vdc
Forward-current transfer ratio	3076	I_B = -0.20 mAdc V_{CE} = -1.0 Vdc		h_{FE}	30	400	---
Collector breakdown voltage	3011	I_C = -25 µAdc V_{BE} = 0.15 Vdc		BV_{CEX}	-18	---	Vdc
Base emitter voltage (saturated)	3066	Test cond. A I_B = -0.20 mAdc I_C = -6.0 mAdc		$V_{BE}(sat)$	---	-0.4	Vdc
Turn-on time		(See figures 2 and 3)		t_{on}	---	0.80	µsec
Turn-off time		(See figures 4 and 5)		t_{off}	---	0.90	µsec

TABLE II. Group B inspection

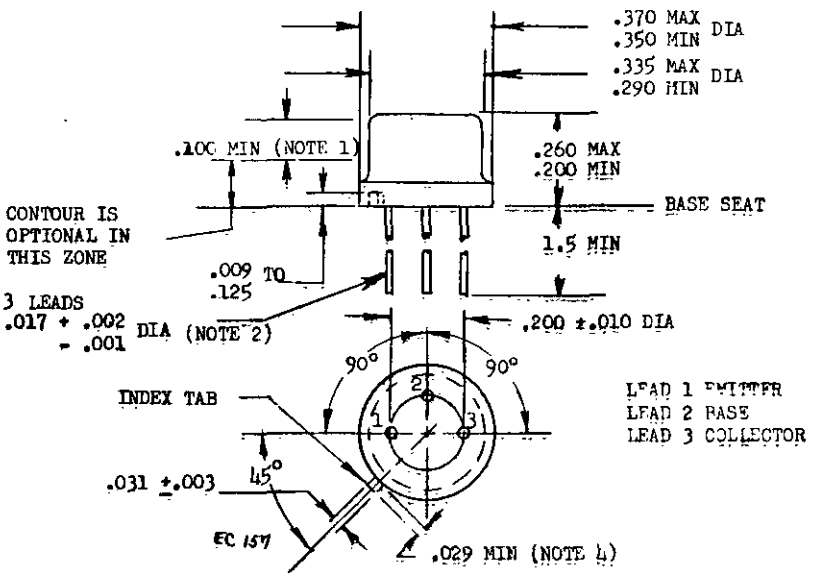
Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 1			20				
Physical dimensions	2066	(See figure 1)		---	---	---	---
Subgroup 2			20				
Solderability	2026			---	---	---	---
Thermal shock (temperature cycling)	1051	Test cond. A T_A = -55°C to +85°C		---	---	---	---
Thermal shock (glass strain)	1056	Test cond. A T_A = 0° to +85°C		---	---	---	---
Moisture resistance	1021			---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -15 Vdc T_A = +60° C		I_{CBO}	---	-49	µAdc
Collector to emitter voltage (saturated)	3071	I_C = -6.0 mAdc I_B = -0.22 mAdc		$V_{CE}(sat)$	---	-0.20	Vdc
Subgroup 3			20				
Shock	2016	Nonoperating; 500 G, approx. 1 msec, 5 blows in each orientation: X_1, Y_1, Y_2, and Z_1.		---	---	---	---
Vibration fatigue	2046	Nonoperating; 10 G		---	---	---	---

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TABLE II. Group B inspection (Continued)

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		
	Method	Details			Min	Max	Unit
Subgroup 3 (Cont.)							
Vibration, variable frequency	2056	Nonoperating		---	---	---	---
Constant acceleration	2006	10,000 G; in each orientation: X_1, Y_1, Y_2, and Z_1.		---	---	---	---
End points:							
(Same as subgroup 2)							
Subgroup 4			20				
Terminal strength (lead fatigue)	2036	Test cond. E		---	---	---	---
Subgroup 5			20				
Salt atmosphere (corrosion)	1041			---	---	---	---
End points:							
(Same as subgroup 2)							
Subgroup 6			λ = 10				
High-temperature life (nonoperating)	1031	T_{stg} = +85° C		---	---	---	---
End points:							
Collector to emitter voltage (saturated)	3071	I_C = -6.0 mAdc I_B = -0.22 mAdc		$V_{CE}(sat)$	---	-0.20	Vdc
Collector to base cutoff current	3036	Bias cond D. V_{CB} = -15 Vdc T_A = +60° C		I_{CBO}	---	-49	µAdc
Collector breakdown voltage	3011	I_C = -35 µAdc V_{BE} = 0.15 Vdc		BV_{CEX}	-17	---	Vdc
Forward-current transfer ratio	3076	I_B = -0.2 mAdc V_{CE} = -1.0 Vdc		h_{FE}	24	---	---
Subgroup 7			λ = 10				
Steady state operation life	1026	T_A = 60° C (See figure 6 for specific conditions)		---	---	---	---
End points:							
(Same as subgroup 6)							

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NOTES:

1. The diameter in this zone is controlled for automatic handling. The total variation in actual diameter from the true diameter, due to taper and out of round within this zone will not exceed 0.010 inch.
2. The specified lead diameter applies in the zone between 0.05 and 0.25 inch from the base seat. Between 0.25 and 1.5 inches a maximum diameter of 0.021 inch is held. Outside of these zones the lead diameter is not controlled.
3. All dimensions in inches.
4. Measured from the maximum diameter of the base.
5. All leads are isolated from the case.

FIGURE 1. Dimensions of transistor, type 2N1853.

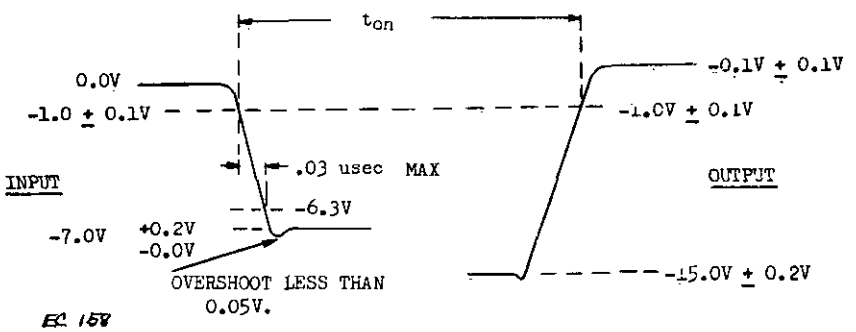
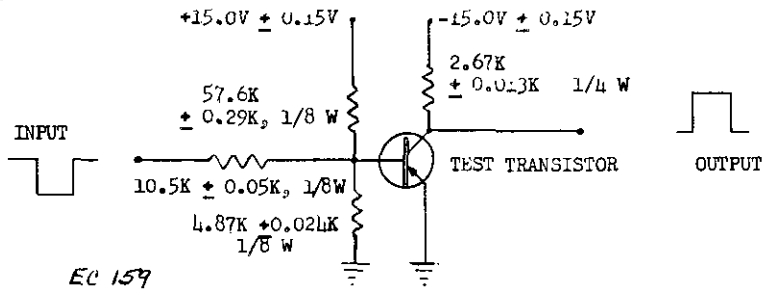


FIGURE 2. Definition of turn-on time, t_{on}.

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NOTE:
The input waveform shall be generated using a mercury-wetted relay (for example, Clare type HGS-1014 or equivalent) which switches between exactly ground and a low-impedance (100 ohm) -7.0V power supply. Voltages shall be checked with a 0.1 percent electronic voltmeter at regular time intervals. A transistor is satisfactory if any (not necessarily every) input voltage setting between -7.0 and -6.8V. results in t_{on} less than, or equal to, the maximum limit.

FIGURE 3. Test circuit for turn-on time.

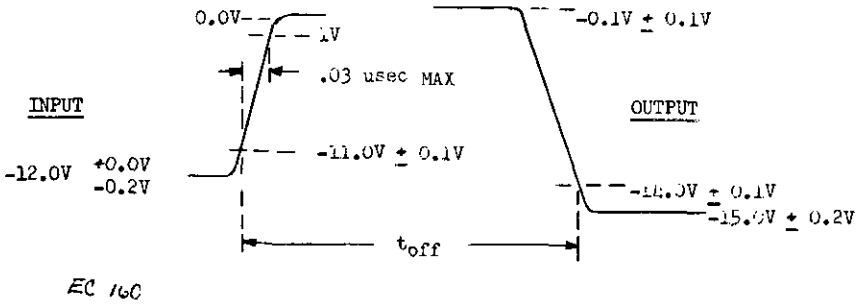
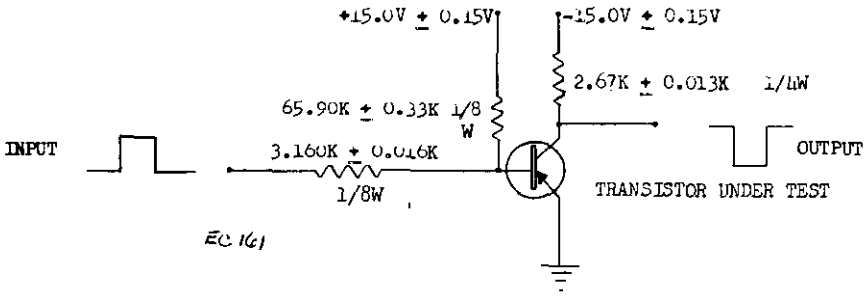


FIGURE 4. Definition of turn-off time t_{off}.



NOTE:
The input waveform shall be generated using a mercury-wetted relay, (for example, Clare type HGS-1014 or equivalent) which switches between exactly ground and a low-impedance (100 ohm) -12.0V power supply. Voltages shall be checked with a 0.1 percent electronic voltmeter at regular time intervals. A transistor is satisfactory if any (not necessarily every) input voltage setting between -12.0 and -12.2V results in t_{off} less than or equal to the maximum limit.

FIGURE 5. Test circuit for turn-off time.

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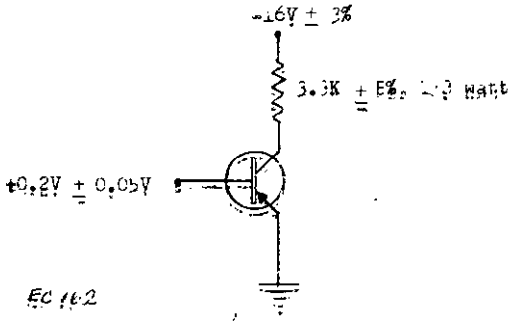


FIGURE 6. Operating life test circuit.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I and II.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

Review activity:
Navy - SH

User activities:
Navy - AS, OS, MC, CG

Preparing activity:
Navy - EC

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